

Statistics

Descriptive and Frequentist Statistics

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Definition:

Statistics is the branch of mathematics concerned with the **collection, organization, analysis, interpretation, and presentation of data** to obtain useful conclusions and support decision-making.

In simple words:

Statistics helps us convert raw data into meaningful information and knowledge for making decisions.

Two Major Areas

- Descriptive Statistics
- Inferential Statistics

Frequentist statistics is one important framework for statistical inference.

Data → Information → Knowledge → Intelligence

- **Data:** Raw facts, observations, measurements or values.
- **Information:** Processed and organized data that has meaning.
- **Knowledge:** Understanding obtained by interpreting information using context, rules and experience.
- **Intelligence:** Ability to use knowledge for reasoning, problem-solving, planning and decision-making.

Example: From Data to Intelligence

Suppose student marks are:

10, 12, 15, 15, 18, 20, 25

- ① **Data:** The raw marks.
- ② **Information:** Average mark, highest mark, lowest mark, etc.
- ③ **Knowledge:** Understanding whether student performance is improving or declining.
- ④ **Intelligence:** Identifying weak areas and recommending suitable topics for additional practice.

Data provides the raw material → Intelligence supports action.

Artificial Intelligence

Definition:

Artificial Intelligence (AI) is a branch of computer science that develops systems capable of performing tasks that normally require human intelligence.

Examples of such activities include:

- Learning
- Reasoning
- Problem-solving
- Planning
- Decision-making
- Perception
- Content generation

AI and Statistics

AI systems use **data and algorithms** to learn patterns, make predictions, reason about problems and perform intelligent tasks.

Humans and AI

Humans	AI Systems
Natural intelligence	Artificial intelligence
Human abilities	Algorithms and computational models
Learn from experience	Learn patterns from data
Use knowledge and experience	Use learned patterns and models
Reason and make decisions	Perform computational reasoning and decision-making
Biological brain	Computer systems

Humans: Abilities + Knowledge

AI: Algorithms + Data

ARTIFICIAL INTELLIGENCE

Think	Plan	Reason	Act
Analyze patterns	Choose steps	Evaluate alternatives	Perform action

Data + Algorithms → Learning → Reasoning → Action

AI does not necessarily possess human intelligence in the same sense as humans; rather, it is designed to perform or simulate specific intelligent activities.

Main Processes of AI

Three important applications of AI are:

1. Decision Making

AI analyzes data and selects an appropriate option or action.

2. Optimization

AI searches for the best or most suitable solution among many possible solutions.

3. Content Generation

AI generates new text, images, audio, video, code and other content based on learned patterns.

Descriptive Statistics

Definition:

Descriptive statistics is concerned with organizing, summarizing and presenting data in a meaningful form.

Major topics include:

- Measures of central tendency
- Measures of dispersion
- Interquartile Range (IQR)
- Box plots
- Outliers
- Histograms
- Skewness
- Kurtosis
- Sample mean and variance
- Order statistics
- Sample covariance

Measures of Central Tendency

Central tendency describes the **central or typical value** of a dataset.

- ① Mean
- ② Median
- ③ Mode

For example:

10, 12, 15, 15, 18

- Mean = 14
- Median = 15
- Mode = 15

Mean

The arithmetic mean is obtained by adding all observations and dividing by the number of observations.

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Example:

$$\bar{x} = \frac{10, 12, 15, 15, 18}{5} = 14$$

Advantages

Uses every observation and is easy to calculate.

Limitation

The mean is sensitive to extreme values or outliers.

Median

The median is the middle value after arranging observations in ascending or descending order.
For an odd number of observations:

10, 12, 15, 18, 20

$$\text{Median} = 15$$

For an even number of observations:

10, 12, 15, 18

$$\text{Median} = \frac{12 + 15}{2} = 13.5$$

The median is less affected by outliers than the mean.

Mode

The mode is the value that occurs most frequently.

Example:

10, 12, 15, 15, 18, 20

Since 15 occurs twice:

$$\text{Mode} = 15$$

A dataset may be:

- Unimodal – one mode
- Bimodal – two modes
- Multimodal – multiple modes
- Without a mode

Measures of Dispersion

Dispersion describes how widely the observations are spread around the center.

Important measures:

- Range
- Variance
- Standard deviation
- Interquartile range

Range

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

Example:

10, 12, 15, 18, 20

$$\text{Range} = 20 - 10 = 10$$

Population Variance

Definition

Population variance measures the average squared deviation of each observation from the population mean.

For a population

$$X = \{x_1, x_2, \dots, x_N\}$$

the population mean is

$$\mu = \frac{1}{N} \sum_{i=1}^N x_i$$

and the population variance is

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

Population Standard Deviation

Definition

Population standard deviation measures the typical spread of observations around the population mean.

The population standard deviation is the positive square root of the population variance:

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$$

Equivalently,

$$\sigma = \sqrt{\sigma^2}$$

Interquartile Range (IQR)

The Interquartile Range measures the spread of the middle 50% of the data.

$$IQR = Q_3 - Q_1$$

where:

- Q_1 = First quartile
- Q_3 = Third quartile

Example:

$$Q_1 = 15, \quad Q_3 = 25$$

Therefore:

$$IQR = 25 - 15 = 10$$

IQR is relatively resistant to extreme observations.

Box Plot

A box plot summarizes data using five important values:

- 1 Minimum
- 2 First quartile (Q_1)
- 3 Median
- 4 Third quartile (Q_3)
- 5 Maximum



The box represents the **IQR** and the lines extending from the box are called **whiskers**.

Outliers

An outlier is an observation that is unusually far from the remaining observations.
Using the IQR method:

$$\text{Lower Fence} = Q_1 - 1.5(IQR)$$

$$\text{Upper Fence} = Q_3 + 1.5(IQR)$$

Observations below the lower fence or above the upper fence are commonly classified as outliers.

Example:

$$Q_1 = 10, \quad Q_3 = 20$$

$$IQR = 10$$

Histogram

A histogram represents the distribution of numerical data.

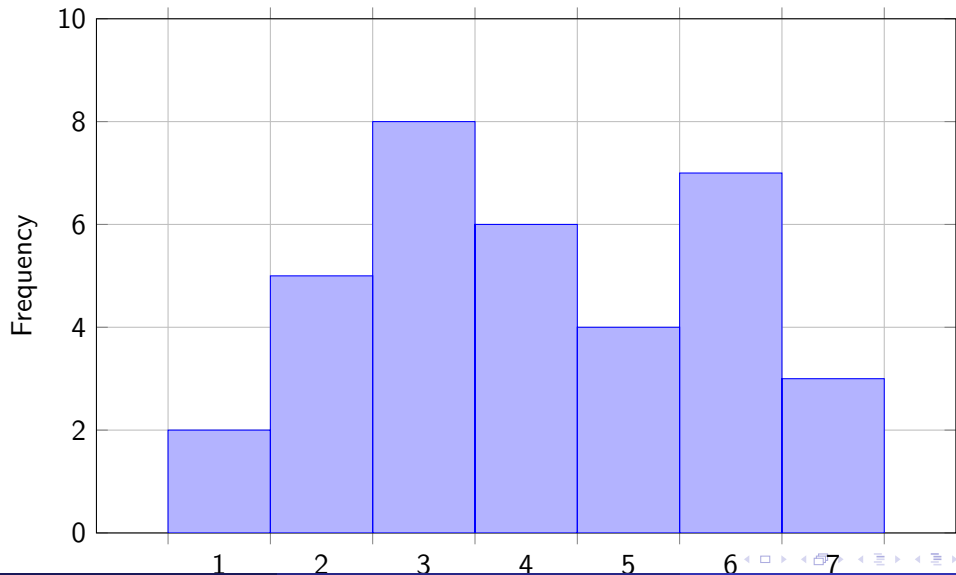
The observations are divided into intervals called **bins**, and the height of each bar represents the frequency.

Bin	Frequency
0–10	3
10–20	7
20–30	5
30–40	2

A histogram helps identify:

- Center
- Spread
- Distribution shape
- Skewness
- Possible outliers
- Number of modes

Histogram



Skewness

Skewness measures the **asymmetry** of a distribution.

Symmetric

Mean $\approx$ Median

Positive / Right Skew

The distribution has a longer right tail.

Typically:

Mean $>$ Median

Negative / Left Skew

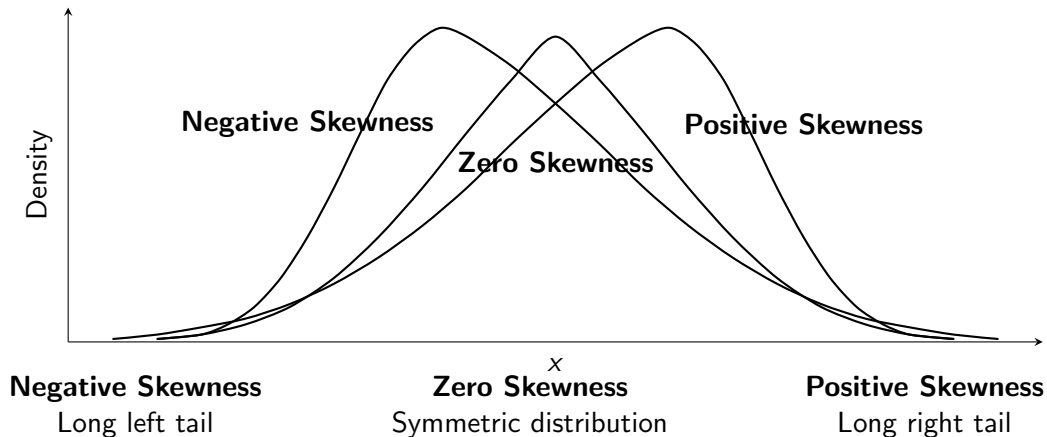
The distribution has a longer left tail.

Typically:

Mean $<$ Median

Skewness

Skewness measures the asymmetry of a probability distribution.



$$\text{Skewness} < 0$$

$$\text{Skewness} = 0$$

$$\text{Skewness} > 0$$

Kurtosis

Kurtosis describes aspects of the **tail heaviness** of a distribution.
One population definition is:

$$\text{Kurtosis} = \frac{E[(X - \mu)^4]}{\sigma^4}$$

For a normal distribution:

$$\text{Kurtosis} = 3$$

Excess kurtosis is:

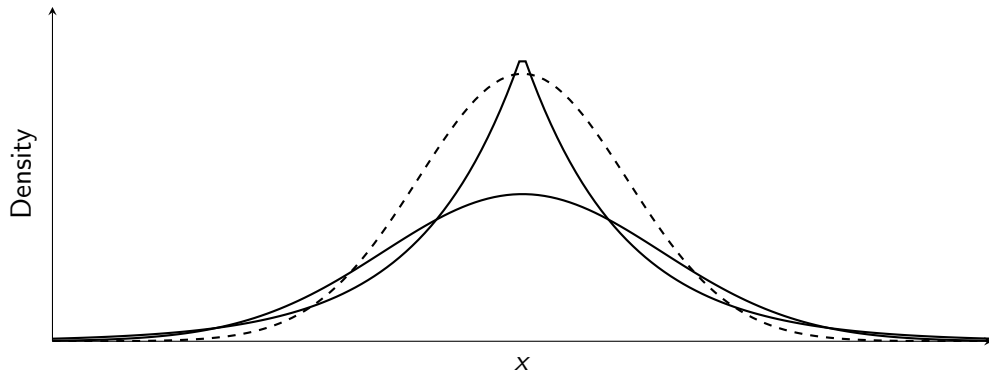
$$\text{Excess Kurtosis} = \text{Kurtosis} - 3$$

Therefore, normal distribution has excess kurtosis:

$$0$$

Kurtosis

Kurtosis describes the heaviness of the tails of a distribution.



— Platykurtic - - - Mesokurtic — Leptokurtic

Platykurtic

Flatter distribution

Mesokurtic

Normal-like distribution

Leptokurtic

More peaked distribution

Sample Mean and Sample Variance

For a sample:

$$X_1, X_2, \dots, X_n$$

the sample mean is:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

The sample variance is:

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

The sample mean estimates the population mean μ and the sample variance estimates the population variance σ^2 .

Order Statistics

Suppose the observations are:

8, 3, 10, 5, 2

After sorting:

2, 3, 5, 8, 10

These ordered observations are called **order statistics**. They are denoted by:

$$X_{(1)}, X_{(2)}, \dots, X_{(n)}$$

where:

$$X_{(1)} = \text{smallest observation}$$

and

$$X_{(n)} = \text{largest observation}$$

Order statistics are used to calculate medians, quartiles, percentiles, minimum and maximum values.

Sample Covariance

Covariance measures the direction of the linear relationship between two variables.
For variables X and Y :

$$S_{XY} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$$

Interpretation

- $S_{XY} > 0$: variables tend to increase together.
- $S_{XY} < 0$: one tends to increase as the other decreases.
- $S_{XY} \approx 0$: little linear association.

Zero covariance does not necessarily imply that there is no relationship; a nonlinear relationship may exist.

Sample Covariance Matrix

For variables:

$$X_1, X_2, \dots, X_p$$

the sample covariance matrix is:

$$S = \begin{bmatrix} S_{11} & S_{12} & \cdots & S_{1p} \\ S_{21} & S_{22} & \cdots & S_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ S_{p1} & S_{p2} & \cdots & S_{pp} \end{bmatrix}$$

where:

$$S_{ii} = \text{Variance of } X_i$$

and

$$S_{ij} = \text{Covariance of } X_i \text{ and } X_j$$

Since:

$$S_{ij} = S_{ji}$$

Example: Covariance Matrix

Consider:

$$S = \begin{bmatrix} 4 & 2 & -1 \\ 2 & 9 & 3 \\ -1 & 3 & 16 \end{bmatrix}$$

Therefore:

- $\text{Var}(X_1) = 4$
- $\text{Var}(X_2) = 9$
- $\text{Var}(X_3) = 16$
- $\text{Cov}(X_1, X_2) = 2$
- $\text{Cov}(X_1, X_3) = -1$
- $\text{Cov}(X_2, X_3) = 3$

Covariance matrices are widely used in:

Machine Learning, PCA, Multivariate Statistics

Frequentist Statistics

Frequentist statistics is a framework for statistical inference based on the behavior of statistical procedures under **repeated sampling**.

In the frequentist framework:

- Population parameters are fixed but unknown.
- Data are treated as random.
- Samples are used to make conclusions about populations.

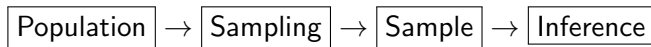
Example:

$$X_1, X_2, \dots, X_n$$

are observations used to estimate the unknown population mean μ .

Sampling

Sampling is the process of selecting observations from a population.



Population: Complete set of objects or individuals of interest.

Sample: A subset selected from the population.

Example:

- Population: All students in a university.
- Sample: 100 students selected for a survey.

Types of Sampling

Simple Random Sampling

Every member has an equal or known chance of selection.

Stratified Sampling

Population is divided into groups called strata and samples are taken from each group.

Systematic Sampling

Every k -th observation is selected after a random starting point.

Cluster Sampling

Population is divided into clusters and some clusters are selected.

Parameter and Statistic

Parameter:

A numerical characteristic of a population.

μ = Population mean

σ^2 = Population variance

Statistic:

A quantity calculated from sample data.

$\bar{X}$ = Sample mean

S^2 = Sample variance

Statistical Estimation

Estimation is the process of using sample data to estimate an unknown population parameter.

Point Estimation

Provides a single value.

$$\hat{\mu} = \bar{X}$$

Interval Estimation

Provides a range of plausible values.

$$L < \mu < U$$

The estimator $\hat{\theta}$ is calculated from sample data and is used to estimate the unknown parameter θ .

Mean Square Error

Mean Square Error (MSE) measures the average squared difference between an estimator and the true parameter.

For estimator $\hat{\theta}$:

$$MSE(\hat{\theta}) = E[(\hat{\theta} - \theta)^2]$$

MSE can be decomposed as:

$$MSE(\hat{\theta}) = \text{Var}(\hat{\theta}) + [\text{Bias}(\hat{\theta})]^2$$

where:

$$\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta$$

A smaller MSE generally indicates an estimator with better overall accuracy under the chosen criterion.

Consistency

An estimator is **consistent** if it approaches the true population parameter as the sample size increases.

For estimator $\hat{\theta}_n$:

$$\hat{\theta}_n \xrightarrow{P} \theta \quad \text{as } n \rightarrow \infty$$

This means:

More data $\rightarrow$ Estimator becomes increasingly close to the true parameter

Under standard conditions, the sample mean is a consistent estimator of the population mean.

Confidence Intervals

A confidence interval is an interval calculated from sample data using a procedure designed to contain an unknown population parameter at a specified long-run frequency.

If population standard deviation σ is known:

$$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

For a 95% confidence interval:

$$\bar{X} \pm 1.96 \frac{\sigma}{\sqrt{n}}$$

If σ is unknown:

$$\bar{X} \pm t_{\alpha/2, n-1} \frac{S}{\sqrt{n}}$$

Interpretation of a 95% Confidence Interval

A 95% confidence procedure means:

If we repeatedly take samples and construct intervals using the same method, approximately 95% of those intervals will contain the true parameter.

It does **not** strictly mean that there is a 95% probability that the fixed population parameter lies inside the particular interval that has already been calculated.

Parametric Model Estimation

A parametric model assumes that the population distribution belongs to a specified family described by a finite number of parameters. Example:

$$X \sim N(\mu, \sigma^2)$$

The parameters are:

$$\theta = (\mu, \sigma^2)$$

They can be estimated using sample data:

$$\hat{\mu} = \bar{X}$$

$$\hat{\sigma}^2 = S^2$$

Common parametric models include:

- Normal distribution
- Binomial distribution
- Poisson distribution
- Exponential distribution

Maximum Likelihood Estimation

Maximum Likelihood Estimation (MLE) is an important method of parametric estimation. For observations:

$$X_1, X_2, \dots, X_n$$

the likelihood function is:

$$L(\theta) = \prod_{i=1}^n f(x_i|\theta)$$

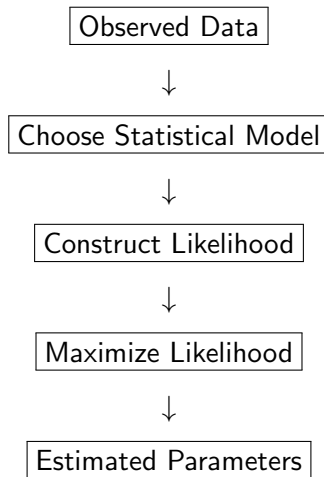
The maximum likelihood estimator is:

$$\hat{\theta} = \arg \max_{\theta} L(\theta)$$

Often the log-likelihood is maximized instead:

$$\hat{\theta} = \arg \max_{\theta} \log L(\theta)$$

MLE: Basic Process



Non-Parametric Model Estimation

Non-parametric methods make fewer assumptions about the exact form of the population distribution.

Instead of assuming:

$$X \sim N(\mu, \sigma^2)$$

the distribution can be estimated more flexibly from the data.

Examples:

- Empirical distribution
- Histogram-based estimation
- Kernel density estimation
- Rank-based methods

Kernel Density Estimation

Kernel Density Estimation (KDE) estimates the probability density function from observations.

$$\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - X_i}{h}\right)$$

where:

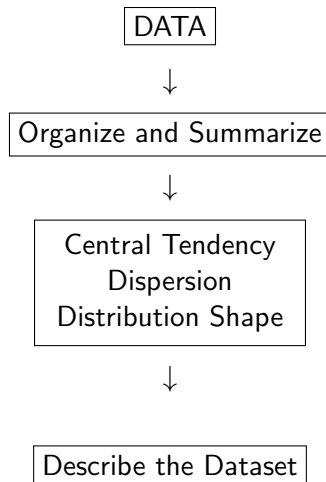
- K = Kernel function
- h = Bandwidth
- n = Number of observations

The bandwidth controls the smoothness of the estimated density.

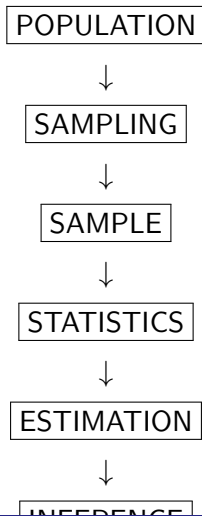
Parametric vs Non-Parametric

Feature	Parametric	Non-Parametric
Assumptions	Specific distributional family	Fewer distributional assumptions
Flexibility	Lower	Higher
Parameters	Finite-dimensional	Not restricted to a fixed finite-dimensional form
Example	Normal model	KDE
Data	Can work well with smaller samples if model is appropriate	Often needs more data for good accuracy

Descriptive Statistics: Overall View



Frequentist Statistics: Overall View



Complete Connection

Data → Information → Knowledge → Intelligence

Statistics → Analyze Data → Extract Information → Support Knowledge → Decision Making

AI combines:

Data + Algorithms + Models → Learning + Reasoning + Decision Making

Applications in AI and Machine Learning

Statistical concepts are fundamental to AI and Machine Learning.

- Mean and variance help understand datasets.
- Covariance helps identify relationships between features.
- Covariance matrices are used in PCA.
- Histograms help understand feature distributions.
- Skewness and kurtosis describe distribution shape.
- Sampling provides training and testing data.
- Estimation is used to learn model parameters.
- MSE is widely used as a loss function in regression.
- Confidence intervals quantify uncertainty.
- Parametric and non-parametric methods provide different approaches to modelling data.

- **Descriptive statistics** summarizes and describes observed data.
- Measures of central tendency describe the center of data.
- Measures of dispersion describe the spread.
- IQR, box plots and histograms help understand distributions and outliers.
- Skewness and kurtosis describe distribution shape.
- Covariance describes relationships between variables.
- **Frequentist statistics** uses samples to make inferences about populations.
- Estimation, MSE, consistency and confidence intervals are important tools of inference.
- Parametric and non-parametric methods provide different approaches to statistical modelling.
- These concepts form an important statistical foundation for AI and Machine Learning.

Thank You